



Al-Kufa University
Engineering College



Electronic And Communication Department

3rd stage

LABORATORY MANUAL

Sequential Logic Circuit Lab

"Laboratory Experiments of Digital Logic Circuits"

Experiment 2

System's Design Using Sequential Logic gates

Case study 1

Prepared by:
Asst.lec. Muhiialdin M.ridha

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1. Aims:

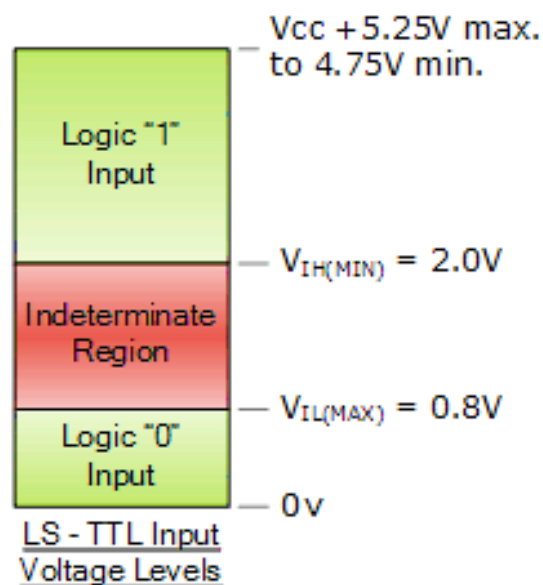
- In this experiment, you will understand how to connect switches to logic gates and how to choose suitable resistor values in your logic circuit design.
- This experiment demonstrates a real world application of D flip flop.

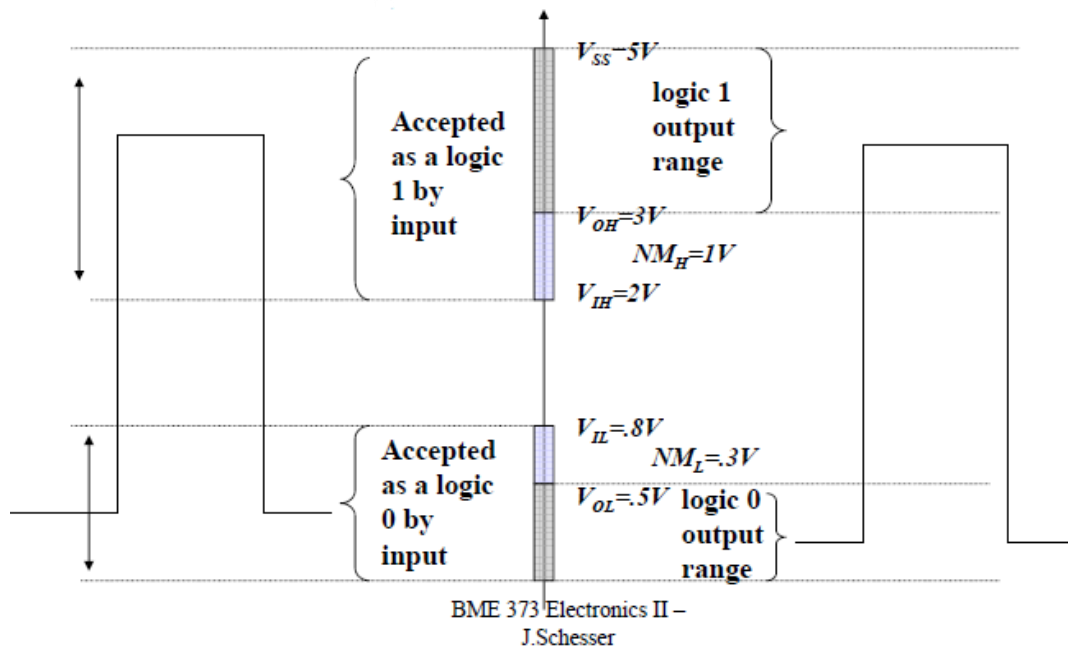
2. Theory :

2.1 Important criteria in designing of any Logic system

For a given logic family, there is a range of output voltage which represents a logic 1 and another range which represents a logic 0. These values can be seen in the datasheet of any logic IC which are:

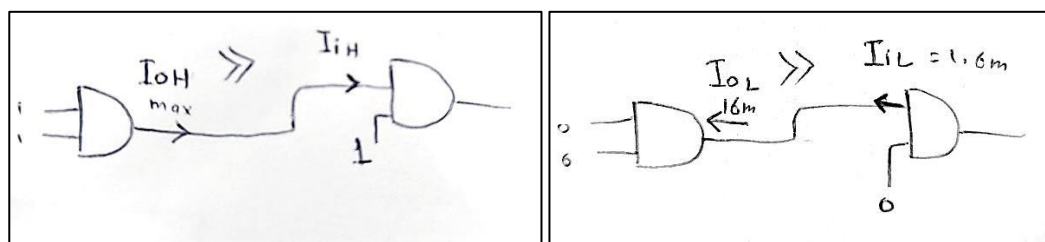
- $V_{iL}(\text{max})$: is the maximum input voltage guaranteed to be recognized as a logic "0" (low) input.
- $V_{iH}(\text{min})$: is the minimum input voltage guaranteed to be recognized as a logic "1" (high) input.
- V_{oL} : is the highest output produced for logic 0 (ideally =0)
- V_{oH} : is the lowest output produced for a logic 1 (ideally = vcc)





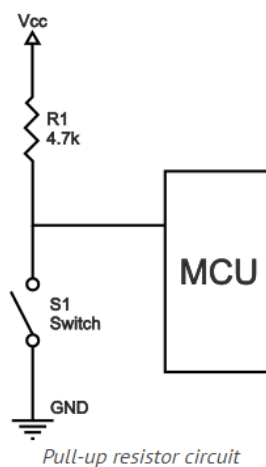
Currents also need to be considered in designing of any logic circuit. There are 4 types of currents need to pay attention to them:

- I_{OH} : The output current supplied by the device (sourced) when the output is high.
- I_{IH} : The maximum input current to support a logic 1.
- I_{OL} : The output current supported by the device (sunked) when the output is low.
- I_{IL} : The maximum input current to support a logic 0.



2.2 Pull up and Pull down resistors

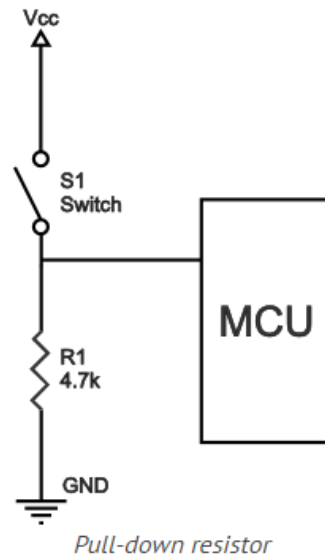
Pull-up resistors are resistors used in logic circuits to ensure a well-defined logical level at a pin under all conditions. As a reminder, digital logic circuits have three logic states: high, low and floating (or high impedance). The high-impedance state occurs when the pin is not pulled to a high or low logic level, but is left “floating” instead. A good illustration of this is an unconnected input pin of a microcontroller. It is neither in a high or low logic state, and the microcontroller might unpredictably interpret the input value as either a logical high or logical low. Pull-up resistors are used to solve the dilemma for the microcontroller by pulling the value to a logical high state, as seen in the follow figure.



$$R_{MAX} = \frac{V_{CC} - V_{IH(MIN)}}{I_{IH}}$$

Ideally we would want a logic “1” to be as close to Vcc as possible to guarantee 100% that the gate see’s a HIGH (logic-1) input through the pull-up resistor. However, we do not want the resistor value to be too low as this would increase current flow into the gate increasing power dissipation.

Pull-down resistors work in the same manner as pull-up resistors, except that they pull the pin to a logical low value. They are connected between ground and the appropriate pin on a device. An example of a pull-down resistor in a digital circuit can be seen in the following figure.



$$R_{MAX} = \frac{V_{IL(MAX)} - 0}{I_{IL}}$$

The pull-down resistor must have a larger resistance than the impedance of the logic circuit, or else it might be able to pull the voltage down by too much and the input voltage at the pin would remain at a constant logical low value – regardless of the switch position

2.3 Case Study:

Digital systems can be implemented using two types of circuits, combinational and sequential circuits. If the component delays are not taken into account as in combinational circuits, that means that the value of their output signals only depends on the values of their input signals at the same time. However, many digital system specifications cannot be implemented by combinational circuits because the value of an output signal could be a function of not only the value of the input signals at the same time, but also the value of the input signals at preceding times.

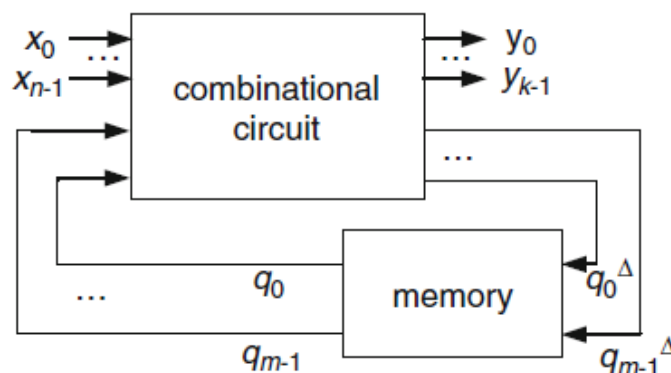


Fig. 1.1 Sequential circuit

consider the circuit that controls a robot vacuum cleaner (the photo of a commercial robot is shown in the Figure below

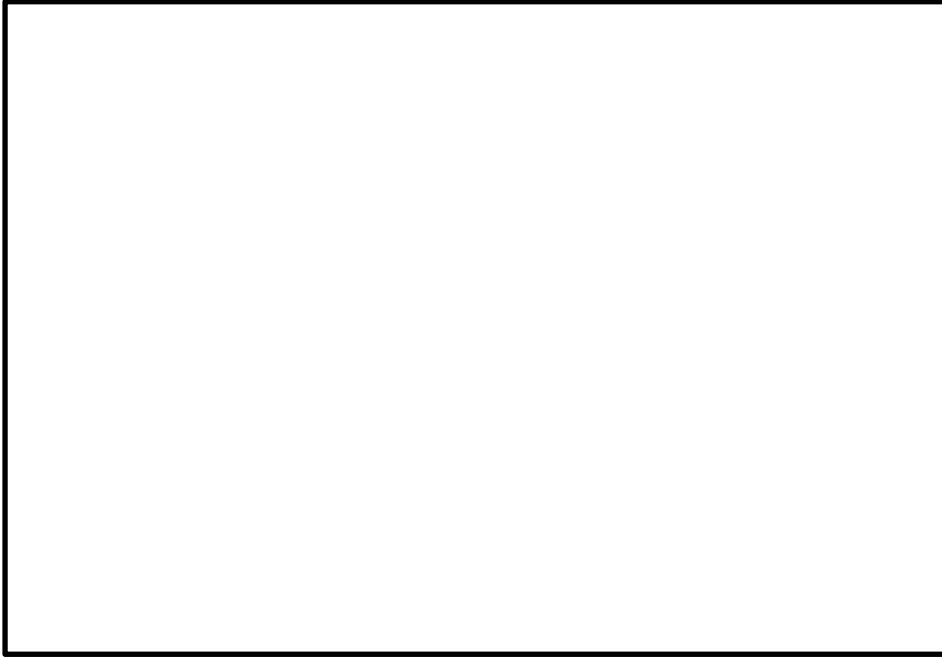


- The robot includes a sensor that generates a binary signal $OB = 1$ when it detects an obstacle in front of it.
- The robot can execute three orders (**outputs**) under the control of two binary inputs: LR (left rotate) , RR(right rotate).
- The **outputs** are as follow:
 - move forward (LR = RR = 0)
 - turn 90° to the left (LR = 1, RR = 0)
 - turn 90° to the right (LR = 0, RR = 1)

The specification of the robot control circuit is the following:

- If there is no obstacle: move forward.
- When an obstacle is detected: turn to the right until there is no more obstacle.
- The next time an obstacle is detected: turn to the left until there is no more obstacle.
- The next time an obstacle is detected: turn to the right until there is no more obstacle, and so on.

This behavior cannot be implemented by a combinational circuit. In order to take a decision it is not enough to know whether there is an obstacle or not; it is necessary to know the latest ordered movements. The state diagram shows all the possible scenarios. Try to do it yourself



The state diagram should fulfill the following rules:

- If the previous command was turn to the right and if there is no obstacle then move forward.
- If the previous command was turn to the right and there is still an obstacle then keep turning to the right.
- If the previous command was turn to the left and if there is no obstacle then move forward.
- If the previous command was turn to the left and there is still an obstacle then keep turning to the left.
- If the previous command was move forward and if there is no obstacle then keep moving forward.
- If the previous command was move forward and there is an obstacle and the latest rotation was to the left then turn to the right.
- If the previous command was move forward and there is an obstacle and the latest rotation was to the right then turn to the left.

This analysis suggests the definition of four internal states: Try to do it

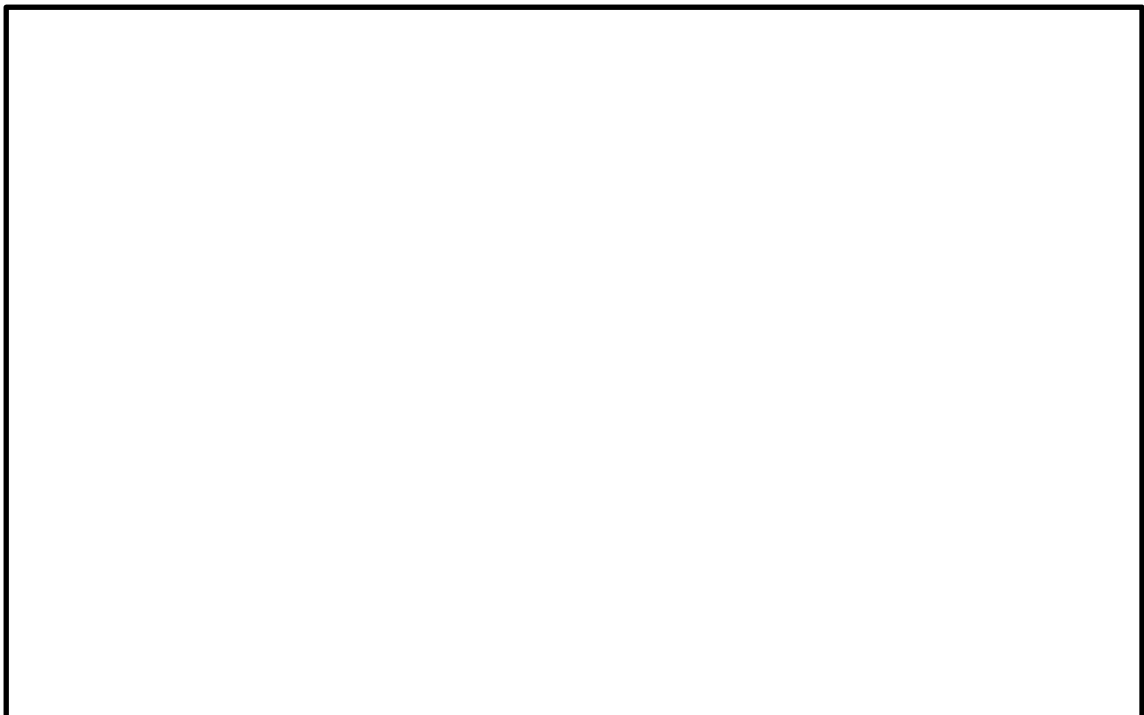
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3. Procedures:

3.1 Procedure 1:

1. connect **HD74LS08P** AND gate IC as you did in the previous experiment.
2. Choose suitable resistor values in your circuit design based on the voltage and current parameters in the data sheet and based on the equations of pull up and pull down resistors.
3. Connect 10kohm variable resistor on the output and change it until no short circuit appears on the dc power supply
4. measure all the input and the output voltages and currents
5. Record all the measured values then draw the circuit diagram clarifying all the resistor values that you used in your design.



3.2 Procedure 2 :

1. to make your circuit design you need first to fill the first 5 columns of the state table shown below. You can do that with the help of the state diagram .

Table 1

Present state		OB	Next state		Flip flop input	
q_0	q_1		q_0'	q_1'	D_0	D_1
0	0					
0	1					
1	0					
1	1					
0	0					
0	1					
1	0					
1	1					

2. Next, you need to choose what kind of flip flop you want to use in your circuit design. For practical and Lab issues D-flip flop has been chosen for you.

3. From the execution table of D-flip flop fill the last two column in the state table.

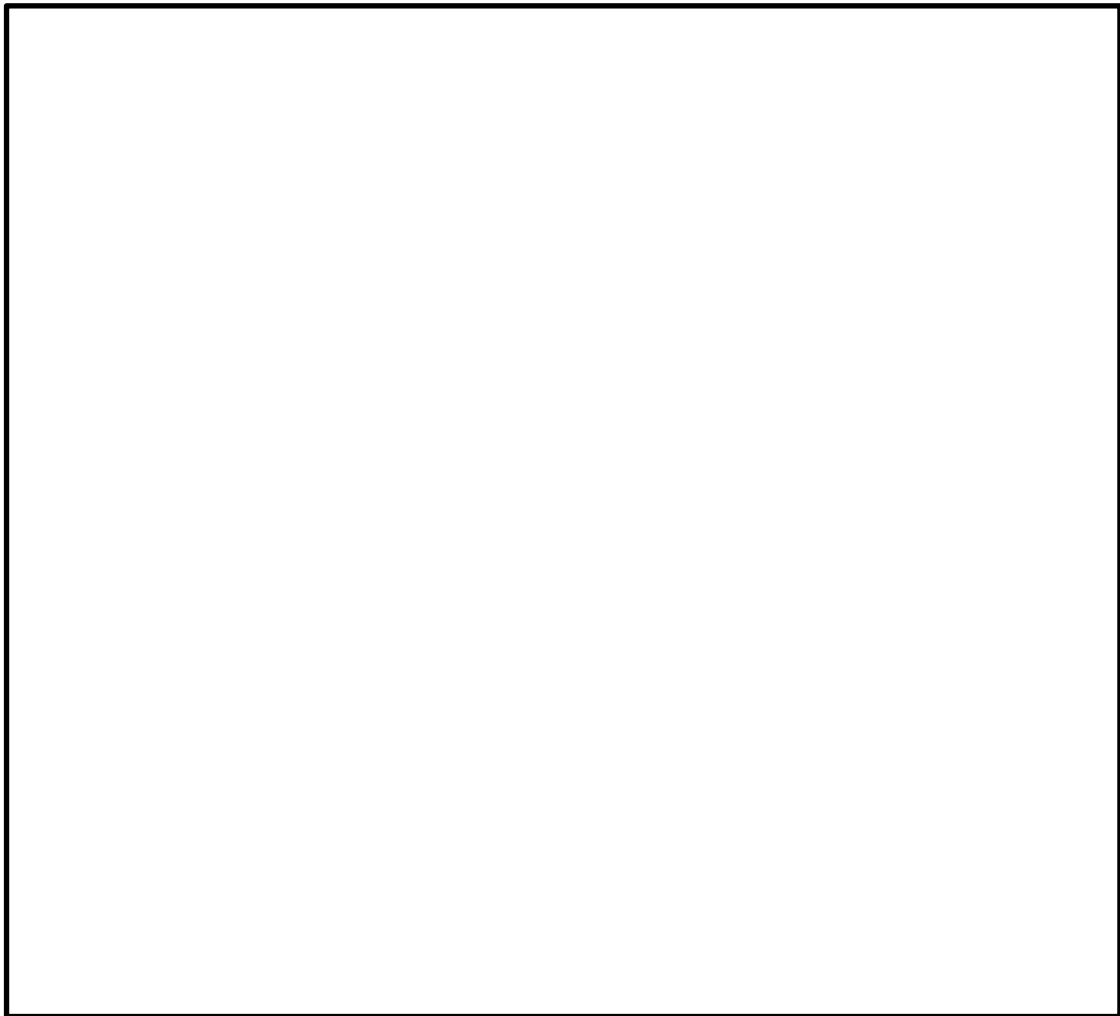
4. Use K-map or De-Morgan theorem to find the Boolean expression for D_0 and D_1

5. From the state diagram find the output values then write down its Boolean expressions.

Table 2

Current state		Output	
q_1	q_0	RR	RL
0	0		
0	1		
1	0		
1	1		

6. From the Boolean expressions of D_0 , D_1 , RR and RL draw the circuit diagram.



7. Use laboratory equipment to connect the circuit practically then compare your practical results with values in table 1 and table 2.